

PITCH DETECTION WITH AVERAGE MAGNITUDE DIFFERENCE FUNCTION USING ADAPTIVE THRESHOLD ALGORITHM FOR ESTIMATING SHIMMER AND JITTER

Hae Young Kim, Jae Sung Lee, Myung-Whun Sung*, Kwang Hyun Kim*, Kwang Suk Park**

Interdisciplinary Program in Medical and Biological Engineering Major,

*Department of Otorhinolaryngology Head and Neck Surgery, College of Medicine,

**Department of Biomedical Engineering, College of Medicine,

Seoul National University, Seoul, Korea

E-mail: kspark@snuvh.snu.ac.kr

Abstract— In order to estimate jitter and shimmer, we suggest the method for the detection of precise pitch and amplitude of voiced speech signal. After low-pass filtering, voiced / unvoiced decision and pitch detection is performed using average magnitude difference function (AMDF). We find minimum value at $t=\tau$ and also check the possible minimum value at $t=\tau/2, \tau/3, \tau/4, \tau/5, \tau/6$. By the application of adaptive threshold, we can estimate pitch more accurately than uniquely applied threshold.

Keywords – jitter, shimmer, pitch detection, average magnitude difference function

I. INTRODUCTION

Jitter and Shimmer estimate phonatory stability as fundamental frequency and amplitude perturbation. As they are very useful and widely-used parameters in clinical situation, it is necessary to estimate pitch and peak-to-peak amplitude of speech signal to estimate jitter and shimmer.

AMDF is one of the conventionally used algorithms which is a variation on the autocorrelation analysis to measure the periodicity of voiced speech. This function has the advantage of sharp resolution of pitch measurement compared with autocorrelation function analysis [1].

We have just estimated a vowel letter, such as a, e, i, o, u. Because complex words or phrases are not necessary for the estimation of jitter and shimmer.

II. METHOD

A block diagram of the jitter and shimmer estimation is given in Fig. 1. The speech is low-pass filtered to have a cutoff frequency of 600 Hz.

We performed voicing decision and pitch detection using AMDF, which is defined as

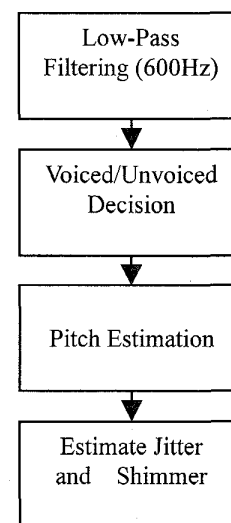


Fig. 1. Jitter and Shimmer estimation process

$$\text{AMDF}(t) = \frac{1}{L} \sum_{i=1}^L |s(i) - s(i-t)| \quad (1)$$

where $s(i)$: the samples of input speech
 $s(i)=[s(1), s(2), \dots, s(L)]$
 $s(i-\tau)$: the samples time shifted

Frame length L was chosen to be 450 samples, corresponding to 22.5ms.

A block diagram of voicing decision process using AMDF is given in Fig. 2. We used the maximum and minimum of AMDF values as a reference for voicing decision. If their ratio is too small, it means that the frame is unvoiced. In addition to voiced and unvoiced region, there may be voiced-to-unvoiced transition region, vice versa. Because there is unvoiced region, we are interested in the voiced region, we classified transition state into unvoiced.

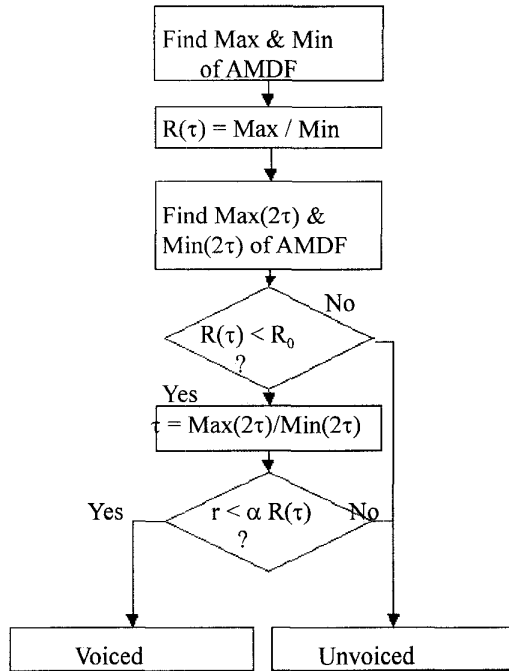


Fig. 2. Voiced / unvoiced decision process

We applied pitch estimation to each voiced region, as follows :

1. We can expect that

$$\tau = n \cdot T_0 \quad (n=1,2,\dots) \quad (2)$$

where τ : shifted time of minimum AMDF
 T_0 : real pitch,

for pitch detection error is almost pitch doubling error [2].

2. We set up threshold (Th), which proportional to the difference between maximum (Max) and minimum (Min) values in the Fig. 3. If AMDF is smaller than threshold around τ , we can regard it as a pitch period. In order to decide n, we defined threshold in each voiced segment :

$$Th = \frac{Max - Min}{N_0} \quad (3)$$

where Max : maximum value in the frame
 Min : minimum value in the frame
 N_0 : constant.

3. If AMDF(t) is smaller than Th in $[\tau/2-10, \tau/2+10]$, corresponding to 1ms around $\tau' = \tau/2$, we regard it as a pitch period and AMDF(τ) as the pitch doubling error.

Similar tests are performed on tripling ($\tau/3$), fourfold ($\tau/4$), fivefold ($\tau/5$), and sixfold ($\tau/6$) errors. We can apply

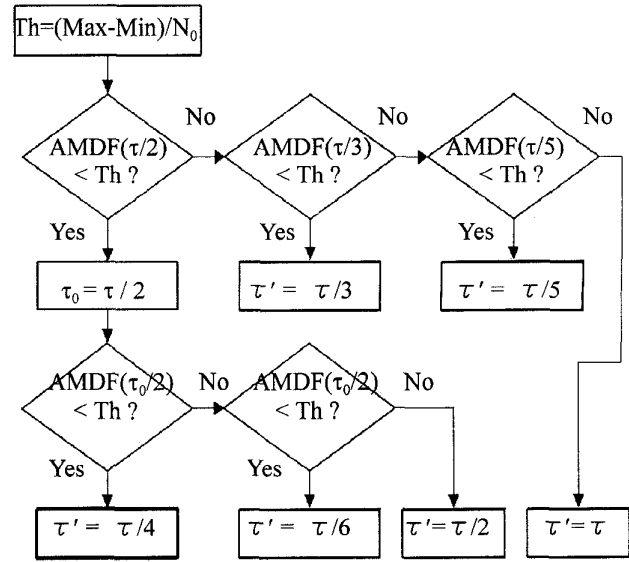


Fig. 3. Pitch detection process

different threshold at each test by changing N_0

4. As sampling rate is 20kHz, fundamental frequency is given by $F_0 = 20000/\tau'$ (Hz)

Jitter and shimmer are defined as,

$$Jitter = \frac{\frac{1}{N-1} \sum_{i=1}^{N-1} |T_0(i) - T_0(i+1)|}{\frac{1}{N} \sum_{i=1}^N T_0(i)} * 100(\%) \quad (4)$$

$$Shimmer = \frac{\frac{1}{N-1} \sum_{i=1}^{N-1} |A(i) - A(i+1)|}{\frac{1}{N} \sum_{i=1}^N A(i)} * 100(\%) \quad (5)$$

where $A(i)$: extracted peak-to-peak amplitude data

$T_0(i)$: extracted pitch period data

N : number of extracted pitch periods

We can get $A(i)$ from the maximum and minimum signal amplitude data around pitch period in i-th frame and $T_0(i) = \tau'$ from the pitch detection algorithm in Fig. 3.

III. RESULT

Fig. 4. shows patterns of voiced sound "a" of a normal female and corresponding AMDF pitch detection result where each vertical line is plotted for τ (Max) and τ' (Pit),

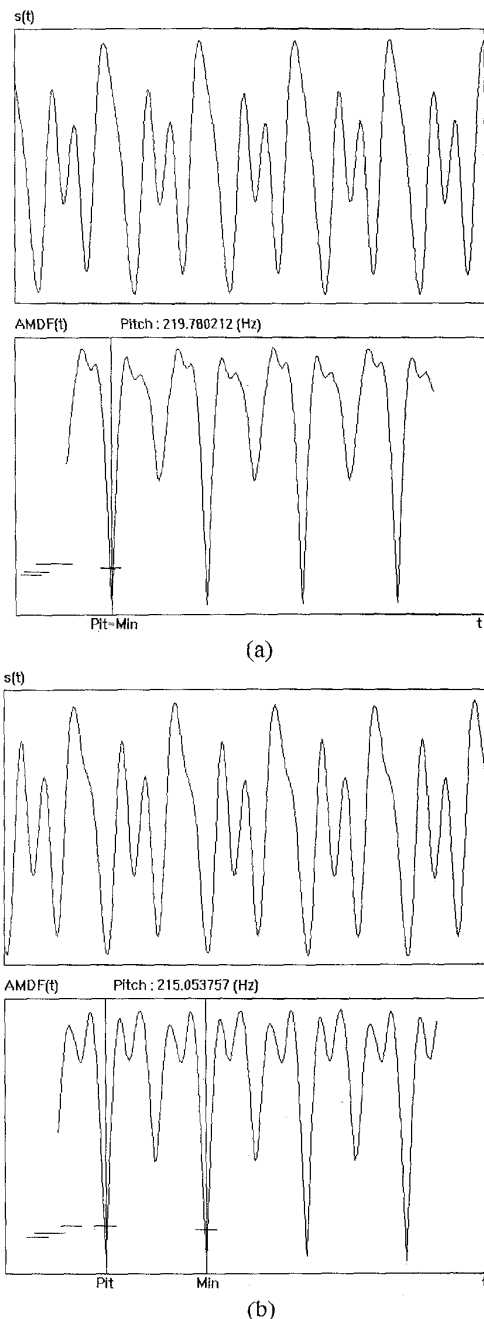


Fig. 4. Voiced speech sound and corresponding AMDF. (a) Pitch is corresponding to τ . (b) Pitch is corresponding to $\tau/2$.

and the horizontal segments above the bottom line are thresholds around τ , $\tau/2$, ..., $\tau/6$.

As you can see, it has correctly detected the first minimum as the pitch of this frame although the lowest

AMDF value is the doubling pitch in Fig. 4(b).

Table I is overall estimated pitches and their average. With these pitches, we've known that jitter is 4.96 % and shimmer is 2.98 % from (4), (5). These values are very small, which is also obvious from Fig. 4., because given data came from a normal female, whose voiced signal is very stable and periodic.

TABLE I
Estimated pitch data using adaptive threshold

Frame	Pitch (Hz)	Frame	Pitch (Hz)
1	217.4	12	217.4
2	217.4	13	215.1
3	219.8	14	217.4
4	219.8	15	217.4
5	219.8	16	217.4
6	219.8	17	217.4
7	217.4	18	217.4
8	217.4	19	217.4
9	219.8	20	219.8
10	219.8	21	219.8
11	217.4		
Average	218.2		

IV. DISCUSSION

We have considered threshold and time simultaneously. It may seem to be unnecessary to estimate pitch doubling errors, for it looks enough to apply threshold and find the first minimum peak [3]. But we must consider many unexpected situations in clinical purpose, so it is more accurate to apply adaptive threshold on each time segment. In Fig. 4, we have chosen N_0 in (3) as 7.27 ($t = \tau/2$, $\tau/3$), 8 ($t = \tau$), 8.89 ($t = \tau/4$, $\tau/5$) and 10 ($t = \tau/6$).

It is necessary to find out optimal N_0 s at each time-segment, that is, how to determine weighting factor on each pitch doubling time.

V. CONCLUSION

We conclude that modified AMDF pitch detection using adaptive threshold estimate pitch more accurately without pitch doubling.

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